

Markov Chains

ORSA Study text



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Markov chains

Introduction

1. Introduction to Markov Chains

- Definition of Markov Chains
- Historical background and key contributors
- Importance and applications in various fields

2. Basic Concepts

- States and state space
- Transition probabilities
- The Markov property (memoryless property)
- Notation and terminology

3. Types of Markov Chains

- Discrete-time Markov Chains (DTMC)
- Continuous-time Markov Chains (CTMC)
- Absorbing Markov Chains
- Hidden Markov Models (HMM)

4. Transition Matrix

- Definition and construction of the transition matrix
- Properties of transition matrices
- Examples of transition matrices

5. Classification of States

- Recurrent vs. transient states
- Absorbing states
- Periodic vs. aperiodic states

6. Long-term Behavior

- Steady-state distributions
- Ergodicity and convergence to equilibrium
- The Limiting distribution
- Applications in predicting long-term outcomes

7. Applications of Markov Chains

- In finance (stock market predictions)
- In computer science (algorithm analysis)
- In biology (population modeling)
- In gaming and decision-making processes

8. Case Studies and Examples

- Real-world examples using Markov Chains
- Detailed case studies highlighting their utility

AUTHOR

Andrey Andreyevich Markov (14 June 1856 – 20 July 1922) was a Russian mathematician best known for his work on stochastic processes. A primary subject of his research later became known as the Markov chain. He was also a strong, close to master-level, chess player.

Markov and his younger brother Vladimir Andreyevich Markov (1871–1897) proved the Markov brothers' inequality. His son, another Andrey Andreyevich Markov (1903–1979), was also a notable mathematician, making contributions to constructive mathematics and recursive function theory.

DIFFERENCES IN COMPARISON WITH OTHER STOCHASTIC PROCESSES

Markov Chains are a specific type of stochastic process characterized by a few distinct features that set them apart from other stochastic processes. Here are the key differences:

1. Memoryless Property

- ***Markov Chains*:** The future state depends only on the current state and not on the sequence of events that preceded it. This is called the Markov property or memoryless property.
- ***Other Stochastic Processes*:** Many other processes, such as ARIMA models or time series, consider historical data and may have dependencies on previous states or observations.

2. State Space

- ***Markov Chains*:** Typically defined over a discrete state space (though there are also continuous Markov processes). The transitions between states occur at discrete time intervals.
- ***Other Stochastic Processes*:** Can operate over both discrete and continuous state spaces. For example, Brownian motion is a continuous stochastic process that does not have a discrete set of states.

3. Transition Probabilities

- ***Markov Chains***: Transition from one state to another is governed by a fixed set of probabilities, represented in a transition matrix. These probabilities remain constant over time in a time-homogeneous Markov Chain.
- ***Other Stochastic Processes***: Transition rules can vary widely. For instance, in a Poisson process, the number of events in a fixed interval is determined by a rate parameter, which can change.

4. Time Parameterization

- ***Markov Chains***: Often analyzed in discrete time steps, where the process is observed at specific intervals (e.g., every hour, day, etc.).
- ***Other Stochastic Processes***: Can be continuous in time, such as diffusion processes that evolve without discrete time steps.

5. Applications

- ***Markov Chains***: Commonly used in scenarios where systems undergo transitions, such as queueing theory, economics, and genetics, where the next state is influenced primarily by the current state.
- ***Other Stochastic Processes***: Used in various fields, such as finance (e.g., modeling stock prices using geometric Brownian motion) or signal processing (e.g., filtering and forecasting).

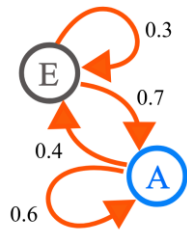
6. Complexity and Structure

- ***Markov Chains***: Generally simpler in structure due to their reliance on the current state alone, making them easier to analyze and compute.
- ***Other Stochastic Processes***: Can be more complex, incorporating various external factors, trends, and seasonality, which require more sophisticated modeling techniques.

In summary, while all Markov Chains are stochastic processes, not all stochastic processes are Markov Chains. The defining feature of Markov Chains is the memoryless property, along with their discrete state transitions and fixed transition probabilities, which differentiate them from other types of stochastic processes.

DIAGRAM AND TRANSITION MATRIX

A following diagram representing a two-state Markov process. The numbers are the probability of changing from one state to another state.



This diagram can be rewritten as a following matrix:

	E	A
E	0.3	0.7
A	0.4	0.6

MARKOV PROPERTY

The Markov property is a fundamental characteristic of Markov processes, which indicates that the future state of a system depends only on its current state and not on the sequence of events that preceded it. In other words, the Markov property embodies the idea of "memorylessness."

Key Aspects of the Markov Property:

1. ***Current State Dependence***: The probability distribution of the next state is conditional only on the current state. This means that if you know the current state, knowing the previous states provides no additional information about the future state.

2. ***Mathematical Representation***: We formulate the Markov Property in mathematical notation as follows:

$$P(X_{t+1} = s \mid X_t = s_t, X_{t-1} = s_{t-1}, \dots, X_0 = s_0) = P(X_{t+1} = s \mid X_t = s_t),$$

for all $t = 1, 2, 3, \dots$ and for all states $s_0, s_1, \dots, s_t, s$.

Definition: Let $\{X_0, X_1, X_2, \dots\}$ be a sequence of discrete random variables. Then

$\{X_0, X_1, X_2, \dots\}$ is a Markov chain if it satisfies the Markov property:

$$P(X_{t+1} = s \mid X_t = s_t, \dots, X_0 = s_0) = P(X_{t+1} = s \mid X_t = s_t),$$

for all $t = 1, 2, 3, \dots$ and for all states $s_0, s_1, \dots, s_t, s$.

3. ***Implications*:** Because of this property, Markov processes can be analyzed using simpler models and methods, making them particularly useful in various fields like statistics, finance, and computer science.

4. ***Types of Markov Processes*:** The Markov property can be applied in different contexts, leading to various types of Markov processes, such as:

- ***Discrete-time Markov Chains (DTMC)*:** Where transitions occur at fixed time intervals.
- ***Continuous-time Markov Chains (CTMC)*:** Where transitions can occur at any point in time.

The Markov property is essential for understanding and working with stochastic processes, as it simplifies the modeling of systems that evolve over time.

APPLICATIONS

Markov Chains have a wide range of applications across various fields due to their ability to model systems that undergo transitions between states. Here are some common real-world applications:

1. ***Finance and Economics***

- ***Stock Market Analysis*:** Markov Chains can be used to model the price movements of stocks, helping in predicting future prices based on current trends.
- ***Credit Scoring*:** Financial institutions utilize Markov Chains to assess the creditworthiness of borrowers by modeling transitions between different credit states (e.g., good credit, default).

2. ***Queueing Theory***

- ***Customer Service*:** In businesses like call centers or banks, Markov Chains are used to model customer arrivals and service processes, helping to optimize staffing and reduce wait times.

3. ***Machine Learning and AI***

- ***Hidden Markov Models (HMM)*:** Used in natural language processing (NLP) for tasks like speech recognition, part-of-speech tagging, and handwriting recognition, where the system states are hidden.
- ***Reinforcement Learning*:** Markov decision processes (MDPs) are foundational to reinforcement learning, where an agent makes decisions based on the current state to maximize rewards.

4. *Genetics and Biology*

- *Population Dynamics*: Markov Chains can model the evolution of populations over generations, analyzing how different traits are passed down.
- *Protein Folding*: Used to predict the states of protein structures as they fold, which is crucial in bioinformatics.

5. *Game Theory and Decision Making*

- *Markov Games*: In strategic games where players make decisions based on the current state, Markov Chains help analyze optimal strategies.
- *Decision Support Systems*: Used in various fields to model decision-making processes where outcomes depend on current conditions.

6. *Weather Prediction*

- *Climate Modeling*: Markov Chains can be utilized to predict weather patterns by modeling transitions between different weather states (e.g., sunny, rainy, cloudy).

7. *Internet and Web Applications*

- *PageRank Algorithm*: Google's PageRank algorithm uses a Markov Chain model to rank web pages based on their link structure, treating web pages as states and links as transitions.
- *User Behavior Modeling*: E-commerce sites use Markov Chains to analyze user navigation patterns, improving recommendations and enhancing user experience.

8. *Social Sciences*

- *Social Mobility*: Markov Chains can model social mobility by analyzing transitions between different socioeconomic statuses over time.

9. *Robotics and Navigation*

- *Path Planning*: In robotics, Markov Chains help in making decisions about navigation and obstacle avoidance in changing environments.

10. *Text Generation and Language Models*

- *Markov Text Generators*: These can create new text based on the statistical properties of existing texts, simulating human-like writing patterns.

These applications demonstrate the versatility and power of Markov Chains in modeling and solving complex problems in various domains.

Certainly! Let's go through a simple example of a discrete-time Markov Chain to illustrate how it works.

EXAMPLE: WEATHER PREDICTION

Imagine we want to model the weather based on two states:

- *Sunny (S)*
- *Rainy (R)*

We can define a Markov Chain with the following transition probabilities:

From/To Sunny (S) Rainy (R)
----- ----- -----
Sunny (S) 0.8 0.2
Rainy (R) 0.4 0.6

From/To	S	R
S	0.8	0.2
R	0.4	0.6

This table tells us:

- If today is sunny, there is an 80% chance that tomorrow will also be sunny and a 20% chance that it will be rainy.
- If today is rainy, there is a 40% chance that tomorrow will be sunny and a 60% chance that it will remain rainy.

Initial State (S_{init})

Let's say the initial state (today's weather) is Sunny. We can represent this as:

- $(P(S_0) = 1)$ (100% probability of being sunny today)
- $(P(R_0) = 0)$ (0% probability of being rainy today)

S _{init}	1	0
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Step-by-Step Transition

We use MS Office Excel, function: MMULT (The **MMULT** function returns the matrix product of two arrays. The result is an array with the same number of rows as array1 and the same number of columns as array2. **Note:** If you have a current version of **Microsoft 365**, then you can simply enter the formula in the top-left-cell of the output range, then press **ENTER** to confirm the formula as a dynamic array formula. Otherwise, the formula must be entered as a legacy array formula by first selecting the output range, entering the formula in the top-left-cell of the output range, and then pressing **CTRL+SHIFT+ENTER** to confirm it. Excel inserts curly brackets at the beginning and end of the formula for you.)

1. *Day 1 (Tomorrow):*

- Probability of Sunny:

$$P(S_1) = P(S_0) \cdot P(S_0 \rightarrow S) + P(R_0) \cdot P(R_0 \rightarrow S) = 1 \cdot 0.8 + 0 \cdot 0.4 = 0.8$$

- Probability of Rainy:

$$P(R_1) = P(S_0) \cdot P(S_0 \rightarrow R) + P(R_0) \cdot P(R_0 \rightarrow R) = 1 \cdot 0.2 + 0 \cdot 0.6 = 0.2$$

S1	0.8	0.2
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2. *Day 2 (The day after tomorrow):*

- Start with the probabilities from Day 1: $P(S_1) = 0.8$ and $P(R_1) = 0.2$.

- Probability of Sunny:

$$P(S_2) = P(S_1) \cdot P(S_1 \rightarrow S) + P(R_1) \cdot P(R_1 \rightarrow S) = 0.8 \cdot 0.8 + 0.2 \cdot 0.4 = 0.64 + 0.08 = 0.72$$

- Probability of Rainy:

$$P(R_2) = P(S_1) \cdot P(S_1 \rightarrow R) + P(R_1) \cdot P(R_1 \rightarrow R) = 0.8 \cdot 0.2 + 0.2 \cdot 0.6 = 0.16 + 0.12 = 0.28$$

S2	0.72	0.28
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This simple example illustrates how Markov Chains allow us to model the transition of states based on current conditions. By using the transition matrix, we can calculate the probabilities of future states based solely on the current state, demonstrating the memoryless property of Markov Chains.